

PRO-cap data processing

This dataset was managed and analyzed using the Resource Management System (<https://github.com/aldenleung/rmsp/>). Raw reads were preprocessed with fastp (v0.23.4) for adapter trimming and unique molecular index (UMI) processing using parameters “*--overlap_len_require 18 --length_required 18*”. The processed reads were aligned to the human reference genome hg38 (GCA_000001405.15) and ribosomal DNA (U13369.1) using STAR (v2.7.11a) with parameters “*--alignMatesGapMax 1000 --outFilterMultimapNmax 10 --outFilterMismatchNmax 1 --outFilterMultimapScoreRange 0 --outSAMattributes AI*”. Uniquely mapped reads were filtered with samtools (v1.18) and deduplicated using umi_tools (v1.1.5). Reads were converted to bigwig and merged between replicates using biodatatools (v0.0.7). Peak calling was performed using PINTS (v1.1.10) with the parameter “*--min-lengths-opposite-peaks 5*”.